

Foundation 4

Logging Component

Foundation Logging Component Developer Reference Manual
First Edition

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Foundation and the Foundation Logging Component written by Dave Batton.

Foundation Logging Component (Fnd_Log)

The Foundation Logging component is an optional Foundation component (it isn't installed in the Foundation Shell by default) that provides a simple method for logging events to an external text file.

The Fnd_Log component is provided at no cost to both Foundation 4 developers and 4D developers that don't use Foundation. The source code for this component is included in the Foundation Construction Set, and it is also included in a separate database included with this component for non-Foundation developers.

This component simply appends text to an external text file. This can be handy when trying to diagnose database problems, especially those that result in the 4D application crashing.

A single method call is used to add text to the log file. The text is added in a tab delimited format, along with the current date and time. The contents of a log file might look something like this:

12/27/2004	19:37:13	On Startup	Database launched.
12/27/2004	19:37:21	Fnd_aa_IO_DisplayTable	Display table called.
12/27/2004	19:37:26	Fnd_aa_IO_DisplayRecord	Display record called.
12/27/2004	19:37:33	On Exit	Database quit.

On Windows, the text file created by this component will be located in the user's Preferences folder. The file will have the same name as the structure file followed by "Log.txt".

On Mac OS X, the text file created by this component will be located in the user's Logs folder, which is located in their Preferences folder. The file will have the same name as the structure file, but with the ".log" file name extension.

On Mac OS 9, the text file created by this component will be located in the user's Preferences folder. The file will have the same name as the structure file followed by "Log". The file type will be set to "TEXT".

The current version of this component does not use or interfere with 4D's **LOG EVENT** command.

This component does not require any other components (not even the Fnd_Gen component).

The Foundation Logging component consists of these routines:

Fnd_Log_AddEntry	Fnd_Log_Info
Fnd_Log_Enable	

Fnd_Log_AddEntry

Fnd_Log_AddEntry (text 1{; text 2..text N})

Parameter	Type	Description
text 1	Text	Text to add to the log
text 2..text N	Text	Additional text to add to the log (optional)

Call this method to add a new entry to the log file. It is safe to call this routine even if we're not currently logging activity.

Fnd_Log_AddEntry ("Database launched.")

If multiple parameters are passed, they will be appended to the log entry separated by tabs.

Because this command can accept multiple values, a very useful technique is to first pass the **Current method name** function, followed by an event description:

```
Fnd_Log_AddEntry (Current method name,"Database launched.")
```

No data will actually be written to the log file until logging is enabled using the *Fnd_Log_Enable* command.

Fnd_Log_Enable

Fnd_Log_Enable ({enable?}) → Boolean

Parameter	Type	Description
enable	Boolean	True to enable logging (optional)
Function result	Boolean	True if logging is enabled

By default, logging is turned off, so calling *Fnd_Log_AddEntry* has no effect. This command allows the developer to turn the logger on or off. Pass **True** to turn on logging, or **False** to turn it off. When logging is enabled, text passed to the *Fnd_Log_AddEntry* command will be written to the log file.

```
Fnd_Log_Enable (True) ` Turn on logging.  
Fnd_Log_AddEntry ("Database launched.")
```

The log file is closed after each entry, so there's no need to turn off logging except to prevent the *Fnd_Log_AddEntry* from writing to the log file.

Fnd_Log_Info

Fnd_Log_Info (info requested) → Text

Parameter	Type	Description
info requested	Text	Info desired
Function Result	Text	Response

Returns the requested information about the component.

```
$version_t:=Fnd_Log_Info ("version")
```

The *Fnd_Log_Info* method will respond to these requests:

Request	Response	Example
name	The component's full name	Foundation Logging
version	The component's version number	4.0.5 beta 2
logging	Either "enabled" or "disabled"	enabled

This component can also be called using the *Fnd_Gen_ComponentInfo* method without first testing to see if the component is installed:

```
$version_t:=Fnd_Gen_ComponentInfo ("Fnd_Log";"version")
```

See the *Fnd_Gen_ComponentInfo* method in the Foundation Developer Reference for more information.